

The Open University of Sri Lanka  
Faculty of Engineering Technology  
Bachelor of Software Engineering Honours  
Academic Year: 2024/2025

EEI4366 - TMA 02

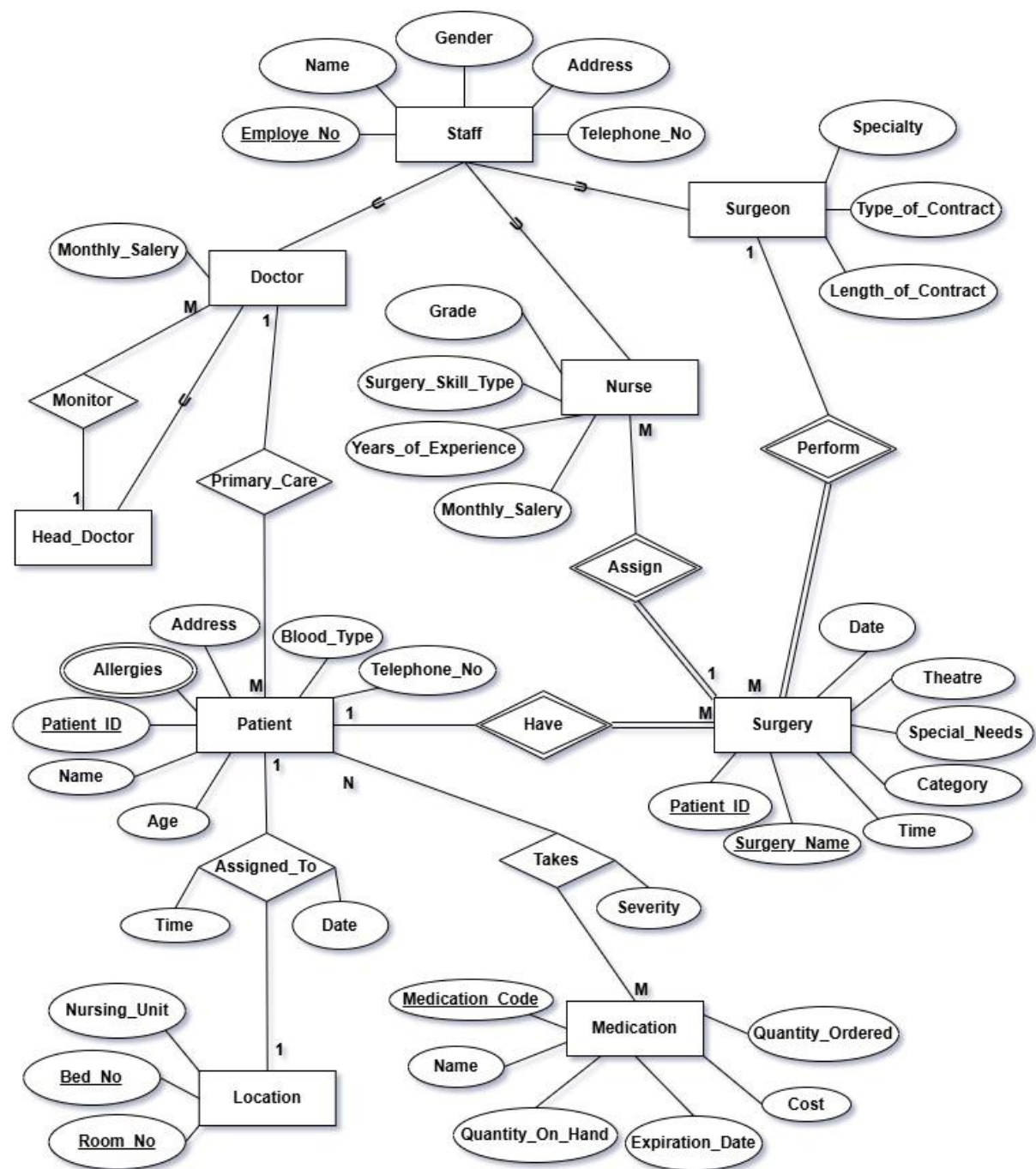
Name: A. P. S. S. Fernando

Registration Number: 523607238

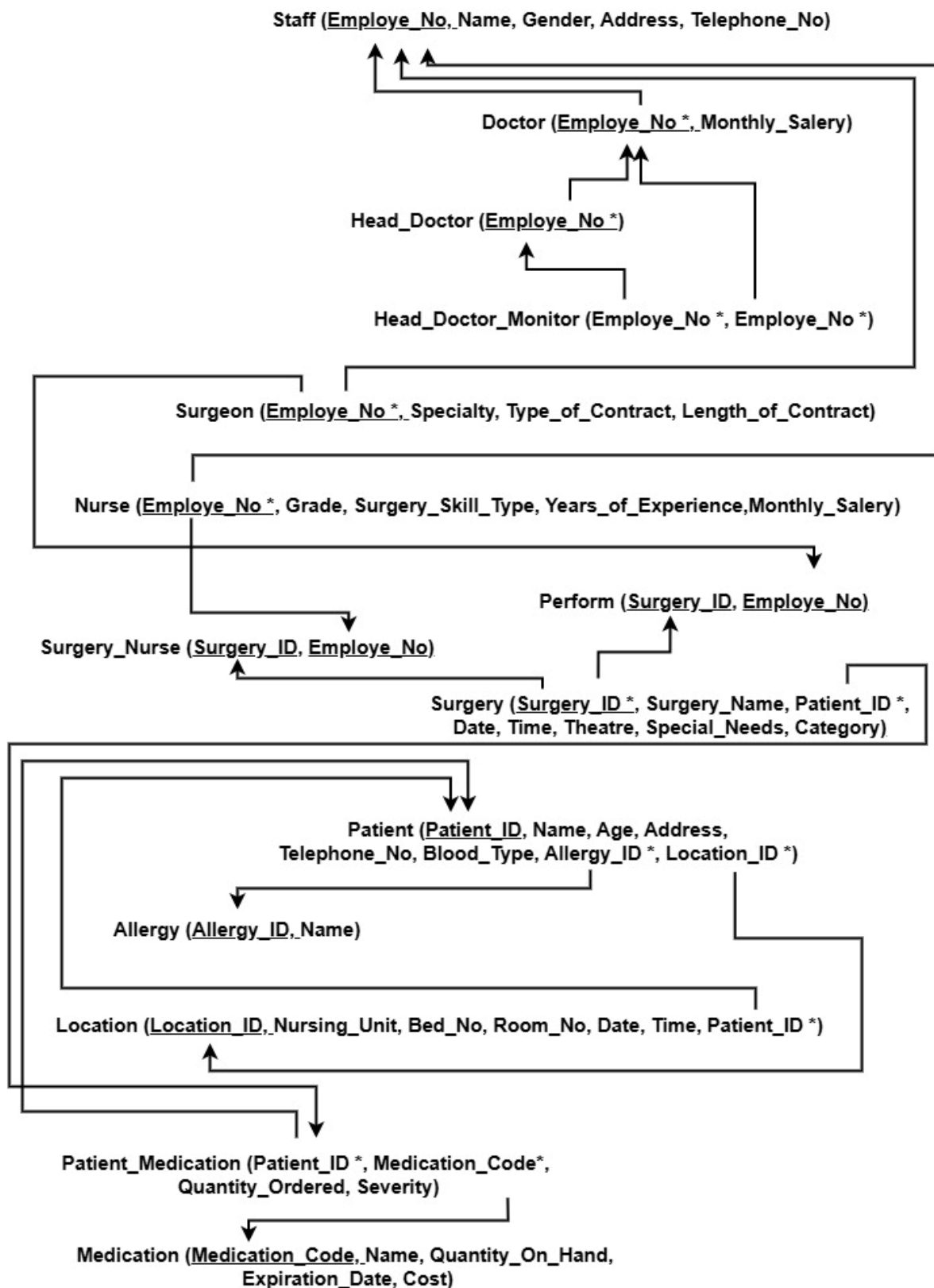
Student Number: s23010373

Submission Date: 05/12/2025

## 1. EER Diagram



## 2. Updated Relational Schema



### 3. TASK 1 - Created Database and Tables

```

1  -- TASK 1
2  • drop database if exists HospitalDB;
3  • create database HospitalDB;
4  • use HospitalDB;
5
6  -- STAFF
7  • create table Staff (
8      Employee_No varchar(10) primary key,
9      Name varchar(100) not null,
10     Gender varchar(10) check (Gender in ('Male','Female')),
11     Address varchar(150),
12     Telephone_No varchar(20) unique,
13     Role enum('Doctor','Nurse','Surgeon') not null
14 );
15
16 -- DOCTOR
17 • create table Doctor (
18     Employee_No varchar(10) primary key,
19     Monthly_Salary decimal(10,2) check (Monthly_Salary > 0),
20     foreign key (Employee_No) references Staff(Employee_No)
21 );
22

```

Output

| # | Time     | Action                                                                                                                                                                                                                                               | Message           | Duration / Fetch |
|---|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| 1 | 20:30:15 | drop database if exists HospitalDB                                                                                                                                                                                                                   | 2 row(s) affected | 0.078 sec        |
| 2 | 20:30:15 | create database HospitalDB                                                                                                                                                                                                                           | 1 row(s) affected | 0.032 sec        |
| 3 | 20:30:15 | use HospitalDB                                                                                                                                                                                                                                       | 0 row(s) affected | 0.000 sec        |
| 4 | 20:30:15 | create table Staff ( Employee_No varchar(10) primary key, Name varchar(100) not null, Gender varchar(10) check (Gender in ('Male','Female')), Address varchar(150), Telephone_No varchar(20) unique, Role enum('Doctor','Nurse','Surgeon') not null) | 0 row(s) affected | 0.109 sec        |
| 5 | 20:30:15 | create table Doctor ( Employee_No varchar(10) primary key, Monthly_Salary decimal(10,2) check (Monthly_Salary > 0), foreign key (Employee_No) references Staff(Employee_No))                                                                         | 0 row(s) affected | 0.062 sec        |

```

23  -- HEAD DOCTOR
24  • create table Head_Doctor (
25     Employee_No varchar(10) primary key,
26     foreign key (Employee_No) references Doctor(Employee_No)
27 );
28
29 -- HEAD DOCTOR MONITORING DOCTOR (weak relation)
30 • create table Head_Doctor_Monitor (
31     HD_Employee_No varchar(10),
32     Doctor_Employee_No varchar(10),
33     primary key (HD_Employee_No, Doctor_Employee_No),
34     foreign key (HD_Employee_No) references Head_Doctor(Employee_No),
35     foreign key (Doctor_Employee_No) references Doctor(Employee_No)
36 );
37
38 -- NURSE
39 • create table Nurse (
40     Employee_No varchar(10) primary key,
41     Grade varchar(50) not null,
42     Surgery_Skill_Type varchar(100),
43     Years_of_Experience int check (Years_of_Experience >= 0),
44     Monthly_Salary decimal(10,2) check (Monthly_Salary > 0),
45     foreign key (Employee_No) references Staff(Employee_No)
46 );

```

Output

| # | Time     | Action                                                                                                                                                                                                                                                                                             | Message           | Duration / Fetch |
|---|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| 5 | 20:30:15 | create table Doctor ( Employee_No varchar(10) primary key, Monthly_Salary decimal(10,2) check (Monthly_Salary > 0), foreign key (Employee_No) references Staff(Employee_No))                                                                                                                       | 0 row(s) affected | 0.062 sec        |
| 6 | 20:34:43 | create table Head_Doctor ( Employee_No varchar(10) primary key, foreign key (Employee_No) references Doctor(Employee_No))                                                                                                                                                                          | 0 row(s) affected | 0.094 sec        |
| 7 | 20:34:43 | create table Head_Doctor_Monitor ( HD_Employee_No varchar(10), Doctor_Employee_No varchar(10), primary key (HD_Employee_No, Doctor_Employee_No), foreign key (HD_Employee_No) references Head_Doctor(Employee_No), foreign key (Doctor_Employee_No) references Doctor(Employee_No))                | 0 row(s) affected | 0.140 sec        |
| 8 | 20:34:44 | create table Nurse ( Employee_No varchar(10) primary key, Grade varchar(50) not null, Surgery_Skill_Type varchar(100), Years_of_Experience int check (Years_of_Experience >= 0), Monthly_Salary decimal(10,2) check (Monthly_Salary > 0), foreign key (Employee_No) references Staff(Employee_No)) | 0 row(s) affected | 0.094 sec        |

```

48 -- SURGEON
49 • create table Surgeon (
50     Employee_No varchar(10) primary key,
51     Specialty varchar(100) not null,
52     Type_of_Contract varchar(100),
53     Length_of_Contract varchar(50),
54     foreign key (Employee_No) references Staff(Employee_No)
55 );
56
57 -- ALLERGY TABLE
58 • create table Allergy (
59     Allergy_ID varchar(10) primary key,
60     Name varchar(100) not null unique
61 );
62
63 -- LOCATION
64 • create table Location (
65     Location_ID varchar(10) primary key,
66     Nursing_Unit varchar(50),
67     Bed_No varchar(10),
68     Room_No varchar(10),
69     Date date not null,
70     Time time not null
71 );

```

Output

| #  | Time     | Action                                                                                                    | Message           | Duration / Fetch |
|----|----------|-----------------------------------------------------------------------------------------------------------|-------------------|------------------|
| 8  | 20:34:44 | create table Nurse ( Employee_No varchar(10) primary key, Grade varchar(50) not null, Surgery_Skill_Ty... | 0 row(s) affected | 0.094 sec        |
| 9  | 20:35:59 | create table Surgeon ( Employee_No varchar(10) primary key, Specialty varchar(100) not null, Type_of_...  | 0 row(s) affected | 0.062 sec        |
| 10 | 20:35:59 | create table Allergy ( Allergy_ID varchar(10) primary key, Name varchar(100) not null unique )            | 0 row(s) affected | 0.125 sec        |
| 11 | 20:35:59 | create table Location ( Location_ID varchar(10) primary key, Nursing_Unit varchar(50), Bed_No varchar...  | 0 row(s) affected | 0.062 sec        |

```

72
73 -- PATIENT
74 • create table Patient (
75     Patient_ID varchar(10) primary key,
76     Name varchar(100) not null,
77     Age int check (Age > 0),
78     Address varchar(150),
79     Telephone_No varchar(20) unique,
80     Blood_Type varchar(10),
81     Allergy_ID varchar(10),
82     Location_ID varchar(10),
83     foreign key (Allergy_ID) references Allergy(Allergy_ID),
84     foreign key (Location_ID) references Location(Location_ID)
85 );
86
87 -- MEDICATION
88 • create table Medication (
89     Medication_Code varchar(10) primary key,
90     Name varchar(100) not null,
91     Quantity_On_Hand int check (Quantity_On_Hand >= 0),
92     Expiration_Date date,
93     Cost decimal(10,2) check (Cost >= 0)
94 );
95

```

Output

| #  | Time     | Action                                                                                                      | Message           | Duration / Fetch |
|----|----------|-------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| 10 | 20:35:59 | create table Allergy ( Allergy_ID varchar(10) primary key, Name varchar(100) not null unique )              | 0 row(s) affected | 0.125 sec        |
| 11 | 20:35:59 | create table Location ( Location_ID varchar(10) primary key, Nursing_Unit varchar(50), Bed_No varchar...    | 0 row(s) affected | 0.062 sec        |
| 12 | 20:37:03 | create table Patient ( Patient_ID varchar(10) primary key, Name varchar(100) not null, Age int check (Ag... | 0 row(s) affected | 0.312 sec        |
| 13 | 20:37:03 | create table Medication ( Medication_Code varchar(10) primary key, Name varchar(100) not null, Quanti...    | 0 row(s) affected | 0.156 sec        |

```

96 -- PATIENT_MEDICATION (M:N)
97 create table Patient_Medication (
98     Patient_ID varchar(10),
99     Medication_Code varchar(10),
100     Severity varchar(50) check (Severity in ('Low','Medium','High')),
101     Quantity_Ordered int check (Quantity_Ordered > 0),
102     primary key (Patient_ID, Medication_Code),
103     foreign key (Patient_ID) references Patient(Patient_ID),
104     foreign key (Medication_Code) references Medication(Medication_Code)
105 );
106
107 -- SURGERY
108 create table Surgery (
109     Surgery_ID varchar(10) primary key,
110     Surgery_Name varchar(100) not null,
111     Patient_ID varchar(10),
112     Date date not null,
113     Time time not null,
114     Theatre varchar(50),
115     Special_Needs varchar(200),
116     Category varchar(50) check (Category in ('Minor','Major','Critical')),
117     foreign key (Patient_ID) references Patient(Patient_ID)
118 );
119

```

Output

| #  | Time     | Action                                                                                                      | Message           | Duration / Fetch |
|----|----------|-------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| 12 | 20:37:03 | create table Patient ( Patient_ID varchar(10) primary key, Name varchar(100) not null, Age int check (Ag... | 0 row(s) affected | 0.312 sec        |
| 13 | 20:37:03 | create table Medication ( Medication_Code varchar(10) primary key, Name varchar(100) not null, Quanti...    | 0 row(s) affected | 0.156 sec        |
| 14 | 20:38:05 | create table Patient_Medication ( Patient_ID varchar(10), Medication_Code varchar(10), Severity varch...    | 0 row(s) affected | 0.188 sec        |
| 15 | 20:38:05 | create table Surgery ( Surgery_ID varchar(10) primary key, Surgery_Name varchar(100) not null, Patient...   | 0 row(s) affected | 0.203 sec        |

```

120 -- PERFORM (SURGEON 1-M SURGERIES)
121 create table Perform (
122     Employee_No varchar(10),
123     Surgery_ID varchar(10),
124     primary key (Employee_No, Surgery_ID),
125     foreign key (Employee_No) references Surgeon(Employee_No),
126     foreign key (Surgery_ID) references Surgery(Surgery_ID)
127 );
128
129 -- SURGERY_NURSE
130 create table Surgery_Nurse (
131     Surgery_ID varchar(10),
132     Employee_No VARCHAR(10),
133     primary key (Surgery_ID, Employee_No),
134     foreign key (Surgery_ID) references Surgery(Surgery_ID),
135     foreign key (Employee_No) references Nurse(Employee_No)
136 );
137

```

Output

| #  | Time     | Action                                                                                                    | Message           | Duration / Fetch |
|----|----------|-----------------------------------------------------------------------------------------------------------|-------------------|------------------|
| 13 | 20:37:03 | create table Medication ( Medication_Code varchar(10) primary key, Name varchar(100) not null, Quanti...  | 0 row(s) affected | 0.156 sec        |
| 14 | 20:38:05 | create table Patient_Medication ( Patient_ID varchar(10), Medication_Code varchar(10), Severity varch...  | 0 row(s) affected | 0.188 sec        |
| 15 | 20:38:05 | create table Surgery ( Surgery_ID varchar(10) primary key, Surgery_Name varchar(100) not null, Patient... | 0 row(s) affected | 0.203 sec        |
| 16 | 20:39:00 | create table Perform ( Employee_No varchar(10), Surgery_ID varchar(10), primary key (Employee_No, S...    | 0 row(s) affected | 0.094 sec        |
| 17 | 20:39:00 | create table Surgery_Nurse ( Surgery_ID varchar(10), Employee_No VARCHAR(10), primary key (Surger...      | 0 row(s) affected | 0.141 sec        |

## 4. TASK 2 - Database Population

```
138 -- TASK 2
139
140 -- STAFF
141 • insert into Staff values
142 ('ST001','Dr. Amal Perera','Male','Colombo','0771234567','Doctor'),
143 ('ST002','Dr. Nadeesha Silva','Female','Kandy','0779876543','Doctor'),
144 ('ST003','Dr. Kasun Jayasuriya','Male','Galle','0714567890','Doctor'),
145 ('ST004','Nurse Sanduni','Female','Kurunegala','0751122334','Nurse'),
146 ('ST005','Nurse Ruwan','Male','Negombo','0709988776','Nurse'),
147 ('ST006','Surgeon Tharindu','Male','Matara','0766655443','Surgeon'),
148 ('ST007','Surgeon Dilki','Female','Jaffna','0728899001','Surgeon');
149
150 -- Show all STAFF records
151 • select * from Staff;
152
```

| Employee_No | Name                 | Gender | Address    | Telephone_No | Role    |
|-------------|----------------------|--------|------------|--------------|---------|
| ST001       | Dr. Amal Perera      | Male   | Colombo    | 0771234567   | Doctor  |
| ST002       | Dr. Nadeesha Silva   | Female | Kandy      | 0779876543   | Doctor  |
| ST003       | Dr. Kasun Jayasuriya | Male   | Galle      | 0714567890   | Doctor  |
| ST004       | Nurse Sanduni        | Female | Kurunegala | 0751122334   | Nurse   |
| ST005       | Nurse Ruwan          | Male   | Negombo    | 0709988776   | Nurse   |
| ST006       | Surgeon Tharindu     | Male   | Matara     | 0766655443   | Surgeon |
| ST007       | Surgeon Dilki        | Female | Jaffna     | 0728899001   | Surgeon |

Staff 1 x

Output

| #  | Time     | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Message                                                | Duration / Fetch      |
|----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 18 | 20:41:37 | insert into Staff values ('ST001','Dr. Amal Perera','Male','Colombo','0771234567','Doctor'), ('ST002','Dr. Nadeesha Silva','Female','Kandy','0779876543','Doctor'), ('ST003','Dr. Kasun Jayasuriya','Male','Galle','0714567890','Doctor'), ('ST004','Nurse Sanduni','Female','Kurunegala','0751122334','Nurse'), ('ST005','Nurse Ruwan','Male','Negombo','0709988776','Nurse'), ('ST006','Surgeon Tharindu','Male','Matara','0766655443','Surgeon'), ('ST007','Surgeon Dilki','Female','Jaffna','0728899001','Surgeon'); | 7 row(s) affected Records: 7 Duplicates: 0 Warnings: 0 | 0.032 sec             |
| 19 | 20:41:37 | select * from Staff LIMIT 0, 1000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7 row(s) returned                                      | 0.000 sec / 0.000 sec |

```
150 -- Show all STAFF records
151 • select * from Staff;
152
153 -- DOCTOR
154 • insert into Doctor values
155 ('ST001',150000.00),
156 ('ST002',145000.00),
157 ('ST003',160000.00);
158
159 -- Show all DOCTOR records
160 • select * from Doctor;
161
```

| Employee_No | Monthly_Salary |
|-------------|----------------|
| ST001       | 150000.00      |
| ST002       | 145000.00      |
| ST003       | 160000.00      |

Doctor 2 x

Output

| #  | Time     | Action                                                                                   | Message                                                | Duration / Fetch      |
|----|----------|------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 20 | 21:01:36 | insert into Doctor values ('ST001',150000.00), ('ST002',145000.00), ('ST003',160000.00); | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0 | 0.062 sec             |
| 21 | 21:01:36 | select * from Doctor LIMIT 0, 1000                                                       | 3 row(s) returned                                      | 0.000 sec / 0.000 sec |

```
162 -- HEAD DOCTOR
163 • insert into Head_Doctor values
164 ('ST001'),
165 ('ST003');
166
167 -- Show all HEAD DOCTOR records
168 • select * from Head_Doctor;
169
```

| Employee_No |
|-------------|
| ST001       |
| ST003       |

Head\_Doctor 3 x

Output

| #  | Time     | Action                                               | Message                                                | Duration / Fetch      |
|----|----------|------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 22 | 21:21:16 | insert into Head_Doctor values ('ST001'), ('ST003'); | 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 | 0.063 sec             |
| 23 | 21:21:16 | select * from Head_Doctor LIMIT 0, 1000              | 2 row(s) returned                                      | 0.000 sec / 0.000 sec |



```

170 -- HEAD DOCTOR MONITOR
171 • insert into Head_Doctor_Monitor values
172   ('ST001','ST002'),
173   ('ST001','ST003'),
174   ('ST003','ST001'),
175   ('ST003','ST002');
176
177 -- Show all HEAD DOCTOR MONITOR records
178 • select * from Head_Doctor_Monitor;

```

| HD_Employee_No | Doctor_Employee_No |
|----------------|--------------------|
| ST003          | ST001              |
| ST001          | ST002              |
| ST003          | ST002              |
| ST001          | ST003              |

Head\_Doctor\_Monitor4 x

Output

| #  | Time     | Action                                                                                                             | Message                                                | Duration / Fetch      |
|----|----------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 24 | 21:23:19 | insert into Head_Doctor_Monitor values ('ST001','ST002'), ('ST001','ST003'), ('ST003','ST001'), ('ST003','ST002'); | 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 | 0.063 sec             |
| 25 | 21:23:19 | select * from Head_Doctor_Monitor LIMIT 0, 1000                                                                    | 4 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

180 -- NURSE
181 • insert into Nurse values
182   ('ST004','Grade A','High',5,85000.00),
183   ('ST005','Grade B','Medium',3,65000.00);
184
185 -- Show all NURSE records
186 • select * from Nurse;
187

```

| Employee_No | Grade   | Surgery_Skill_Type | Years_of_Experience | Monthly_Salary |
|-------------|---------|--------------------|---------------------|----------------|
| ST004       | Grade A | High               | 5                   | 85000.00       |
| ST005       | Grade B | Medium             | 3                   | 65000.00       |

Nurse 5 x

Output

| #  | Time     | Action                                                                                                   | Message                                                | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 26 | 21:24:06 | insert into Nurse values ('ST004','Grade A','High',5,85000.00), ('ST005','Grade B','Medium',3,65000.00); | 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 | 0.047 sec             |
| 27 | 21:24:06 | select * from Nurse LIMIT 0, 1000                                                                        | 2 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

188 -- SURGEON
189 • insert into Surgeon values
190   ('ST006','Cardiac Surgery','Full-time','2 years'),
191   ('ST007','Neuro Surgery','Contract','1 year');
192
193 -- Show all SURGEON records
194 • select * from Surgeon;
195

```

| Employee_No | Specialty       | Type_of_Contract | Length_of_Contract |
|-------------|-----------------|------------------|--------------------|
| ST006       | Cardiac Surgery | Full-time        | 2 years            |
| ST007       | Neuro Surgery   | Contract         | 1 year             |

Surgeon 6 x

Output

| #  | Time     | Action                                                                                                                       | Message                                                | Duration / Fetch      |
|----|----------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 28 | 21:27:55 | insert into Surgeon values ('ST006','Cardiac Surgery','Full-time','2 years'), ('ST007','Neuro Surgery','Contract','1 year'); | 2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0 | 0.047 sec             |
| 29 | 21:27:55 | select * from Surgeon LIMIT 0, 1000                                                                                          | 2 row(s) returned                                      | 0.000 sec / 0.000 sec |



```

196 -- ALLERGY
197 • insert into Allergy values
198   ('AL001','None'),
199   ('AL002','Peanuts'),
200   ('AL003','Dust'),
201   ('AL004','Seafood'),
202   ('AL005','Pollen');
203
204 -- Show all ALLERGY records
205 • select * from Allergy;

```

| Result Grid                                                                                                                                                                                                                                                                                                                | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |       |      |       |         |       |        |       |         |  |  |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------|----------------|--------------------|-------|------|-------|---------|-------|--------|-------|---------|--|--|--|--|
| <table border="1"> <thead> <tr> <th>Allergy_ID</th> <th>Name</th> </tr> </thead> <tbody> <tr> <td>AL003</td> <td>Dust</td> </tr> <tr> <td>AL001</td> <td>None</td> </tr> <tr> <td>AL002</td> <td>Peanuts</td> </tr> <tr> <td>AL005</td> <td>Pollen</td> </tr> <tr> <td>AL004</td> <td>Seafood</td> </tr> </tbody> </table> | Allergy_ID   | Name  | AL003          | Dust               | AL001 | None | AL002 | Peanuts | AL005 | Pollen | AL004 | Seafood |  |  |  |  |
| Allergy_ID                                                                                                                                                                                                                                                                                                                 | Name         |       |                |                    |       |      |       |         |       |        |       |         |  |  |  |  |
| AL003                                                                                                                                                                                                                                                                                                                      | Dust         |       |                |                    |       |      |       |         |       |        |       |         |  |  |  |  |
| AL001                                                                                                                                                                                                                                                                                                                      | None         |       |                |                    |       |      |       |         |       |        |       |         |  |  |  |  |
| AL002                                                                                                                                                                                                                                                                                                                      | Peanuts      |       |                |                    |       |      |       |         |       |        |       |         |  |  |  |  |
| AL005                                                                                                                                                                                                                                                                                                                      | Pollen       |       |                |                    |       |      |       |         |       |        |       |         |  |  |  |  |
| AL004                                                                                                                                                                                                                                                                                                                      | Seafood      |       |                |                    |       |      |       |         |       |        |       |         |  |  |  |  |

Output

| #  | Time     | Action                                                                                                           | Message                                                | Duration / Fetch      |
|----|----------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 30 | 21:28:47 | insert into Allergy values (AL001:'None'), (AL002:'Peanuts'), (AL003:'Dust'), (AL004:'Seafood'), (AL005:'Poll... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.047 sec             |
| 31 | 21:28:47 | select * from Allergy LIMIT 0, 1000                                                                              | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

207 -- LOCATION
208 • insert into Location values
209   ('L001','North Wing','B12','R101','2025-01-20','10:00:00'),
210   ('L002','South Wing','B09','R102','2025-01-22','09:30:00'),
211   ('L003','East Wing','B05','R201','2025-01-25','08:15:00'),
212   ('L004','West Wing','B14','R103','2025-01-28','11:45:00'),
213   ('L005','North Wing','B07','R202','2025-02-01','14:10:00');
214
215 -- Show all LOCATION records
216 • select * from Location;
217

```

| Result Grid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Filter Rows: | Edit:        | Export/Import: | Wrap Cell Content: |          |      |      |            |     |      |            |          |      |            |     |      |            |          |      |           |     |      |            |          |      |           |     |      |            |          |      |            |     |      |            |          |  |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|----------------|--------------------|----------|------|------|------------|-----|------|------------|----------|------|------------|-----|------|------------|----------|------|-----------|-----|------|------------|----------|------|-----------|-----|------|------------|----------|------|------------|-----|------|------------|----------|--|--|--|--|
| <table border="1"> <thead> <tr> <th>Location_ID</th> <th>Nursing_Unit</th> <th>Bed_No</th> <th>Room_No</th> <th>Date</th> <th>Time</th> </tr> </thead> <tbody> <tr> <td>L001</td> <td>North Wing</td> <td>B12</td> <td>R101</td> <td>2025-01-20</td> <td>10:00:00</td> </tr> <tr> <td>L002</td> <td>South Wing</td> <td>B09</td> <td>R102</td> <td>2025-01-22</td> <td>09:30:00</td> </tr> <tr> <td>L003</td> <td>East Wing</td> <td>B05</td> <td>R201</td> <td>2025-01-25</td> <td>08:15:00</td> </tr> <tr> <td>L004</td> <td>West Wing</td> <td>B14</td> <td>R103</td> <td>2025-01-28</td> <td>11:45:00</td> </tr> <tr> <td>L005</td> <td>North Wing</td> <td>B07</td> <td>R202</td> <td>2025-02-01</td> <td>14:10:00</td> </tr> </tbody> </table> | Location_ID  | Nursing_Unit | Bed_No         | Room_No            | Date     | Time | L001 | North Wing | B12 | R101 | 2025-01-20 | 10:00:00 | L002 | South Wing | B09 | R102 | 2025-01-22 | 09:30:00 | L003 | East Wing | B05 | R201 | 2025-01-25 | 08:15:00 | L004 | West Wing | B14 | R103 | 2025-01-28 | 11:45:00 | L005 | North Wing | B07 | R202 | 2025-02-01 | 14:10:00 |  |  |  |  |
| Location_ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Nursing_Unit | Bed_No       | Room_No        | Date               | Time     |      |      |            |     |      |            |          |      |            |     |      |            |          |      |           |     |      |            |          |      |           |     |      |            |          |      |            |     |      |            |          |  |  |  |  |
| L001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | North Wing   | B12          | R101           | 2025-01-20         | 10:00:00 |      |      |            |     |      |            |          |      |            |     |      |            |          |      |           |     |      |            |          |      |           |     |      |            |          |      |            |     |      |            |          |  |  |  |  |
| L002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | South Wing   | B09          | R102           | 2025-01-22         | 09:30:00 |      |      |            |     |      |            |          |      |            |     |      |            |          |      |           |     |      |            |          |      |           |     |      |            |          |      |            |     |      |            |          |  |  |  |  |
| L003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | East Wing    | B05          | R201           | 2025-01-25         | 08:15:00 |      |      |            |     |      |            |          |      |            |     |      |            |          |      |           |     |      |            |          |      |           |     |      |            |          |      |            |     |      |            |          |  |  |  |  |
| L004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | West Wing    | B14          | R103           | 2025-01-28         | 11:45:00 |      |      |            |     |      |            |          |      |            |     |      |            |          |      |           |     |      |            |          |      |           |     |      |            |          |      |            |     |      |            |          |  |  |  |  |
| L005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | North Wing   | B07          | R202           | 2025-02-01         | 14:10:00 |      |      |            |     |      |            |          |      |            |     |      |            |          |      |           |     |      |            |          |      |           |     |      |            |          |      |            |     |      |            |          |  |  |  |  |

Output

| #  | Time     | Action                                                                                                            | Message                                                | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 32 | 21:29:30 | insert into Location values (L001:'North Wing','B12','R101','2025-01-20','10:00:00'), (L002:'South Wing','B09'... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.062 sec             |
| 33 | 21:29:30 | select * from Location LIMIT 0, 1000                                                                              | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

218 -- PATIENT
219 • insert into Patient values
220   ('PT001','Sahan Perera',25,'Colombo','0771112223','A+','AL001','L001'),
221   ('PT002','Nadeesha Kumari',30,'Kandy','0772223334','B+','AL003','L002'),
222   ('PT003','Milan Jayasuriya',40,'Galle','0713334445','O-','AL002','L003'),
223   ('PT004','Ayesha Fernando',28,'Negombo','0704445556','AB+','AL005','L004'),
224   ('PT005','Thilina Dissanayake',22,'Matara','0755556667','A-','AL004','L005');
225
226 -- Show all PATIENT records
227 • select * from Patient;

```

| Result Grid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Filter Rows:        | Edit: | Export/Import: | Wrap Cell Content: |              |            |             |             |       |              |    |         |            |    |       |      |       |                 |    |       |            |    |       |      |       |                  |    |       |            |    |       |      |       |                 |    |         |            |     |       |      |       |                     |    |        |            |    |       |      |  |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------|----------------|--------------------|--------------|------------|-------------|-------------|-------|--------------|----|---------|------------|----|-------|------|-------|-----------------|----|-------|------------|----|-------|------|-------|------------------|----|-------|------------|----|-------|------|-------|-----------------|----|---------|------------|-----|-------|------|-------|---------------------|----|--------|------------|----|-------|------|--|--|--|--|
| <table border="1"> <thead> <tr> <th>Patient_ID</th> <th>Name</th> <th>Age</th> <th>Address</th> <th>Telephone_No</th> <th>Blood_Type</th> <th>Allergy_ID</th> <th>Location_ID</th> </tr> </thead> <tbody> <tr> <td>PT001</td> <td>Sahan Perera</td> <td>25</td> <td>Colombo</td> <td>0771112223</td> <td>A+</td> <td>AL001</td> <td>L001</td> </tr> <tr> <td>PT002</td> <td>Nadeesha Kumari</td> <td>30</td> <td>Kandy</td> <td>0772223334</td> <td>B+</td> <td>AL003</td> <td>L002</td> </tr> <tr> <td>PT003</td> <td>Milan Jayasuriya</td> <td>40</td> <td>Galle</td> <td>0713334445</td> <td>O-</td> <td>AL002</td> <td>L003</td> </tr> <tr> <td>PT004</td> <td>Ayesha Fernando</td> <td>28</td> <td>Negombo</td> <td>0704445556</td> <td>AB+</td> <td>AL005</td> <td>L004</td> </tr> <tr> <td>PT005</td> <td>Thilina Dissanayake</td> <td>22</td> <td>Matara</td> <td>0755556667</td> <td>A-</td> <td>AL004</td> <td>L005</td> </tr> </tbody> </table> | Patient_ID          | Name  | Age            | Address            | Telephone_No | Blood_Type | Allergy_ID  | Location_ID | PT001 | Sahan Perera | 25 | Colombo | 0771112223 | A+ | AL001 | L001 | PT002 | Nadeesha Kumari | 30 | Kandy | 0772223334 | B+ | AL003 | L002 | PT003 | Milan Jayasuriya | 40 | Galle | 0713334445 | O- | AL002 | L003 | PT004 | Ayesha Fernando | 28 | Negombo | 0704445556 | AB+ | AL005 | L004 | PT005 | Thilina Dissanayake | 22 | Matara | 0755556667 | A- | AL004 | L005 |  |  |  |  |
| Patient_ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Name                | Age   | Address        | Telephone_No       | Blood_Type   | Allergy_ID | Location_ID |             |       |              |    |         |            |    |       |      |       |                 |    |       |            |    |       |      |       |                  |    |       |            |    |       |      |       |                 |    |         |            |     |       |      |       |                     |    |        |            |    |       |      |  |  |  |  |
| PT001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sahan Perera        | 25    | Colombo        | 0771112223         | A+           | AL001      | L001        |             |       |              |    |         |            |    |       |      |       |                 |    |       |            |    |       |      |       |                  |    |       |            |    |       |      |       |                 |    |         |            |     |       |      |       |                     |    |        |            |    |       |      |  |  |  |  |
| PT002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Nadeesha Kumari     | 30    | Kandy          | 0772223334         | B+           | AL003      | L002        |             |       |              |    |         |            |    |       |      |       |                 |    |       |            |    |       |      |       |                  |    |       |            |    |       |      |       |                 |    |         |            |     |       |      |       |                     |    |        |            |    |       |      |  |  |  |  |
| PT003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Milan Jayasuriya    | 40    | Galle          | 0713334445         | O-           | AL002      | L003        |             |       |              |    |         |            |    |       |      |       |                 |    |       |            |    |       |      |       |                  |    |       |            |    |       |      |       |                 |    |         |            |     |       |      |       |                     |    |        |            |    |       |      |  |  |  |  |
| PT004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Ayesha Fernando     | 28    | Negombo        | 0704445556         | AB+          | AL005      | L004        |             |       |              |    |         |            |    |       |      |       |                 |    |       |            |    |       |      |       |                  |    |       |            |    |       |      |       |                 |    |         |            |     |       |      |       |                     |    |        |            |    |       |      |  |  |  |  |
| PT005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Thilina Dissanayake | 22    | Matara         | 0755556667         | A-           | AL004      | L005        |             |       |              |    |         |            |    |       |      |       |                 |    |       |            |    |       |      |       |                  |    |       |            |    |       |      |       |                 |    |         |            |     |       |      |       |                     |    |        |            |    |       |      |  |  |  |  |

Output

| #  | Time     | Action                                                                                                         | Message                                                | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 34 | 21:30:26 | insert into Patient values (PT001:'Sahan Perera',25,'Colombo','0771112223','A+','AL001','L001'), (PT002:'Na... | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.078 sec             |
| 35 | 21:30:27 | select * from Patient LIMIT 0, 1000                                                                            | 5 row(s) returned                                      | 0.016 sec / 0.000 sec |

```

229 -- MEDICATION
230 • insert into Medication values
231 ('MD001','Amoxicillin',200,'2026-01-10',50.00),
232 ('MD002','Paracetamol',500,'2027-05-12',10.00),
233 ('MD003','Cetirizine',300,'2026-11-05',15.00),
234 ('MD004','Insulin',100,'2026-03-30',120.00),
235 ('MD005','Aspirin',450,'2027-02-21',30.00);
236
237 -- Show all MEDICATION records
238 • select * from Medication;
239

```

| Medication_Code | Name        | Quantity_On_Hand | Expiration_Date | Cost   |
|-----------------|-------------|------------------|-----------------|--------|
| MD001           | Amoxicillin | 200              | 2026-01-10      | 50.00  |
| MD002           | Paracetamol | 500              | 2027-05-12      | 10.00  |
| MD003           | Cetirizine  | 300              | 2026-11-05      | 15.00  |
| MD004           | Insulin     | 100              | 2026-03-30      | 120.00 |
| MD005           | Aspirin     | 450              | 2027-02-21      | 30.00  |

Medication 10 x

Output

Action Output

| #  | Time     | Action                                                                                                                                                                                                                                                      | Message                                                | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 36 | 21:31:18 | insert into Medication values (MD001,'Amoxicillin',200,'2026-01-10',50.00), (MD002,'Paracetamol',500,'2027-05-12',10.00), (MD003,'Cetirizine',300,'2026-11-05',15.00), (MD004,'Insulin',100,'2026-03-30',120.00), (MD005,'Aspirin',450,'2027-02-21',30.00); | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.047 sec             |
| 37 | 21:31:18 | select * from Medication LIMIT 0, 1000                                                                                                                                                                                                                      | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

240 -- PATIENT_MEDICATION
241 • insert into Patient_Medication values
242 ('PT001','MD001','High',2),
243 ('PT002','MD003','Medium',1),
244 ('PT003','MD005','Low',3),
245 ('PT004','MD002','Medium',1),
246 ('PT005','MD004','High',2);
247
248 -- Show all PATIENT_MEDICATION records
249 • select * from Patient_Medication;
250

```

| Patient_ID | Medication_Code | Severity | Quantity_Ordered |
|------------|-----------------|----------|------------------|
| PT001      | MD001           | High     | 2                |
| PT002      | MD003           | Medium   | 1                |
| PT003      | MD005           | Low      | 3                |
| PT004      | MD002           | Medium   | 1                |
| PT005      | MD004           | High     | 2                |

Patient\_Medication 11 x

Output

Action Output

| #  | Time     | Action                                                                                                                                                                     | Message                                                | Duration / Fetch      |
|----|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 38 | 21:32:24 | insert into Patient_Medication values (PT001,'MD001','High',2), (PT002,'MD003','Medium',1), (PT003,'MD005','Low',3), (PT004,'MD002','Medium',1), (PT005,'MD004','High',2); | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.063 sec             |
| 39 | 21:32:24 | select * from Patient_Medication LIMIT 0, 1000                                                                                                                             | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

251 -- SURGERY
252 • insert into Surgery values
253 ('SG001','Heart Bypass','PT001','2025-03-01','09:00:00','T1','Ventilator','Critical'),
254 ('SG002','Brain Tumor Removal','PT002','2025-03-03','10:30:00','T2','ICU','Critical'),
255 ('SG003','Bone Fixation','PT003','2025-03-05','11:15:00','T3','X-Ray','Major'),
256 ('SG004','Skin Graft','PT004','2025-03-07','12:45:00','T4','Extra Lighting','Minor'),
257 ('SG005','Appendix Removal','PT005','2025-03-09','08:20:00','T5','Basic Tools','Major');
258
259 -- Show all SURGERY records
260 • select * from Surgery;

```

| Surgery_ID | Surgery_Name        | Patient_ID | Date       | Time     | Theatre | Special_Needs  | Category |
|------------|---------------------|------------|------------|----------|---------|----------------|----------|
| SG001      | Heart Bypass        | PT001      | 2025-03-01 | 09:00:00 | T1      | Ventilator     | Critical |
| SG002      | Brain Tumor Removal | PT002      | 2025-03-03 | 10:30:00 | T2      | ICU            | Critical |
| SG003      | Bone Fixation       | PT003      | 2025-03-05 | 11:15:00 | T3      | X-Ray          | Major    |
| SG004      | Skin Graft          | PT004      | 2025-03-07 | 12:45:00 | T4      | Extra Lighting | Minor    |
| SG005      | Appendix Removal    | PT005      | 2025-03-09 | 08:20:00 | T5      | Basic Tools    | Major    |

Surgery 12 x

Output

Action Output

| #  | Time     | Action                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Message                                                | Duration / Fetch      |
|----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 40 | 21:33:15 | insert into Surgery values (SG001,'Heart Bypass','PT001','2025-03-01','09:00:00','T1','Ventilator','Critical'), (SG002,'Brain Tumor Removal','PT002','2025-03-03','10:30:00','T2','ICU','Critical'), (SG003,'Bone Fixation','PT003','2025-03-05','11:15:00','T3','X-Ray','Major'), (SG004,'Skin Graft','PT004','2025-03-07','12:45:00','T4','Extra Lighting','Minor'), (SG005,'Appendix Removal','PT005','2025-03-09','08:20:00','T5','Basic Tools','Major'); | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.047 sec             |
| 41 | 21:33:15 | select * from Surgery LIMIT 0, 1000                                                                                                                                                                                                                                                                                                                                                                                                                           | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

262 -- PERFORM
263 • insert into Perform values
264   ('ST006','SG001'),
265   ('ST007','SG002'),
266   ('ST006','SG003'),
267   ('ST007','SG004'),
268   ('ST006','SG005');
269
270 -- Show all PERFORM records
271 • select * from Perform;
272

```

| Employee_No | Surgery_ID |
|-------------|------------|
| ST006       | SG001      |
| ST007       | SG002      |
| ST006       | SG003      |
| ST007       | SG004      |
| ST006       | SG005      |

Perform 13 x

Output

| #  | Time     | Action                                                                                                | Message                                                | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 42 | 21:33:58 | insert into Perform values (ST006;SG001), (ST007;SG002), (ST006;SG003), (ST007;SG004), (ST006;SG005); | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.047 sec             |
| 43 | 21:33:58 | select * from Perform LIMIT 0, 1000                                                                   | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

```

273 -- SURGERY_NURSE
274 • insert into Surgery_Nurse values
275   ('SG001','ST004'),
276   ('SG002','ST005'),
277   ('SG003','ST004'),
278   ('SG004','ST005'),
279   ('SG005','ST004');
280
281 -- Show all SURGERY_NURSE records
282 • select * from Surgery_Nurse;

```

| Surgery_ID | Employee_No |
|------------|-------------|
| SG001      | ST004       |
| SG003      | ST004       |
| SG005      | ST004       |
| SG002      | ST005       |
| SG004      | ST005       |

Surgery\_Nurse 14 x

Output

| #  | Time     | Action                                                                                                      | Message                                                | Duration / Fetch      |
|----|----------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------|
| 44 | 21:34:42 | insert into Surgery_Nurse values (SG001;ST004), (SG002;ST005), (SG003;ST004), (SG004;ST005), (SG005;ST004); | 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 | 0.062 sec             |
| 45 | 21:34:42 | select * from Surgery_Nurse LIMIT 0, 1000                                                                   | 5 row(s) returned                                      | 0.000 sec / 0.000 sec |

\*\*\*\*\*